Inorganic CUME
Sept-30-2016

(Note: Answer Any Five. Only Five Will Be Graded)

1. Answer **two** mini questions below, and briefly discuss your chemistry for supporting each answer. (***20 points***)
   i) Why is the bond angle in NH_3 different than (or same as) that in PH_3?
   ii) Which one, among CH_3^- and NH_2^-, has a stronger proton affinity than the other?
   iii) Ranking the hydrated cations by their size: Li^+ (aq), K^+ (aq), Cs^+ (aq) and Na^+ (aq).
   iv) Rank the LiF and CsI in the order of increasing the solubility in water.

2. Based on the Latimer Diagram below,

$$\begin{array}{lcccccccccc} E^o(V) & & 0.564 & & 1.44 & & -0.70 & & 1.51 & & -1.18 \\ & MnO_4^- & \rightarrow & MnO_4^{2-} & \rightarrow & MnO_2 & \rightarrow & Mn^{3+} & \rightarrow & Mn^{2+} & \rightarrow & Mn \end{array}$$

   Explain which Mn species can be used for making Cl_2 from Cl^-, and why so. (***20 points***)
   $\{E^o=1.36\ (V)$ *for* $\frac{1}{2}\ Cl_2 + e^- = Cl^-\}$

3. Use a simple sketch to indicate – (***20 points***)
   i) a (221) lattice plane in an *fcc* crystal unit cell;
   ii) the typical coordination number for each atom in a typical *fcc* crystal;

4. Plotting the electrical resistivity against the temperature for a metal, for an insulator, for a semiconductor, and for a superconductor, respectively. (***20 points***)

5. Explain why a single-sheet of $(BN)_n$ can (or cannot) be as electrically conductive as a graphene (a single-layer of graphite). (***20 points***)

6. Use Fe^{2+} and ligands of your choice to make a diamagnetic O_h complex, and figure out the LFSE and # of unpaired electron(s) for the complex. (***20 points***)

7. Figure out the lowest energy term in the ground state atomic spectrum for the Ni in $Ni(CO)_4$. (***20 points***)

8. Propose a simple reaction that can stabilize the Cp_2Co, and briefly explain why so. (***20 points***)

9. Find a molecule isoelectronic to borazine. Assign a point group for the borazine. (***20 points***)

10. Propose a stable organometallic complex that could form on the graphene, and another one on the $(BN)_n$. Show your reasoning to support your proposal. (***20 points***)